THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 1: **Exploration: Entering the World of Chemistry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In science, a model is best described as: 1

(a) an untested guess
(b) an exact copy of the real object
(c) a simplified representation of a system that keeps what matters for the question
(d) a final truth that never changes

2. A rough estimate is useful when: 1

(a) an approximate answer is enough for a decision
(b) no data are available at all
(c) we want to avoid reasoning
(d) an exact value is required

3. A 'law' of science and a 'theory' differ because: 1

(a) they are the same
(b) a law describes what happens while a theory explains why
(c) a theory is untested
(d) a law is a guess

4. Chemists draw atoms as spheres and bonds as sticks. This is an example of: 1

(a) a model **(b)** a law **(c)** an estimate **(d)** a unit

5. One breath is about 0.5 litre and a person takes about 20000 breaths a day. The air breathed in a day is about: 1

(a) 10000 litres **(b)** 100000 litres **(c)** 100 litres **(d)** 1000 litres

6. The conservation of energy is an example of: 1

(a) a unit
(b) a symbol
(c) an estimate
(d) a general principle that applies in every process

7. A scientific law usually: 1

(a) describes a regular pattern observed in nature
(b) explains why the pattern exists in every detail
(c) cannot be tested
(d) is a personal opinion

8. The gap between a prediction and observation is an opportunity to: 1

(a) change the data **(b)** blame the instrument
(c) stop experimenting **(d)** refine the model

9. An example of a chemical change is: 1

(a) tearing paper **(b)** burning of magnesium
(c) dissolving sugar **(d)** melting of ice

10. A well-tested scientific theory: 1

(a) is a measurement
(b) is just a guess
(c) explains many observations and makes testable predictions
(d) can never be changed

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is a model in science? Give one example from chemistry. 2

12. Why is ignoring air resistance not a mistake when studying a falling ball? 2

13. What do scientists do when a prediction does not match an observation? 2

14. Differentiate between a scientific law and a theory. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain with an example how a model is built and why assumptions are made. 3

16. How does the incident of the aircraft running out of fuel show the importance of units? 3

17. Calculate the mass of 2.5 litres of water in kg and in g (density 1 kg per litre). 3

OR

Describe how an estimate of the number of litres of air breathed in a day can be made. 3

18. Explain how Meghnad Saha's work on stars shows the use of models and laws. 3

OR

Explain the difference between accuracy and precision with an example. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the characteristics of the scientific method with an example of an investigation, including question, hypothesis, experiment, observation and conclusion. 5

OR

Explain, with examples, why measurements are never perfectly exact, and how errors can be reduced. 5

20. Discuss the role of models in science with three examples from chemistry or physics, and explain how models change. 5

OR

Discuss the use of chemistry in everyday life with five examples and the responsibility of scientists. 5

END OF QUESTION PAPER — Total: 40 Marks